Does DVOL add information beyond HAR?

A retrospective BTC variance forecast study | Yuvraj Singh | 26 October 2026

Abstract. Does option-implied variance improve Bitcoin's direct 30-day variance forecasts beyond a heterogeneous autoregressive (HAR) benchmark? I first compare HAR, HAR plus Deribit's option-implied volatility index (DVOL) and raw DVOL on 1,357 historical origins, then evaluate a declared set of 17 specifications on a separate 1,176-origin common cohort. Longer estimation windows lower HAR + DVOL's mean quasi-likelihood (QLIKE) loss. Log-HAR plus DVOL with training-only smearing has the lowest point loss in the broader comparison, while the focal QLIKE-fitted HAR + DVOL versus QLIKE-fitted HAR bootstrap interval includes zero. The data do not separate estimation noise, model misspecification and changes in forecast scale. Inference remains a limitation: the declared subsampling alternative fails the synthetic size checks, and the model confidence set does not establish a winner. The next step is a separately declared prospective comparison of matched positive-variance models; the existing year remains best-effort evidence collection without a formal significance test.

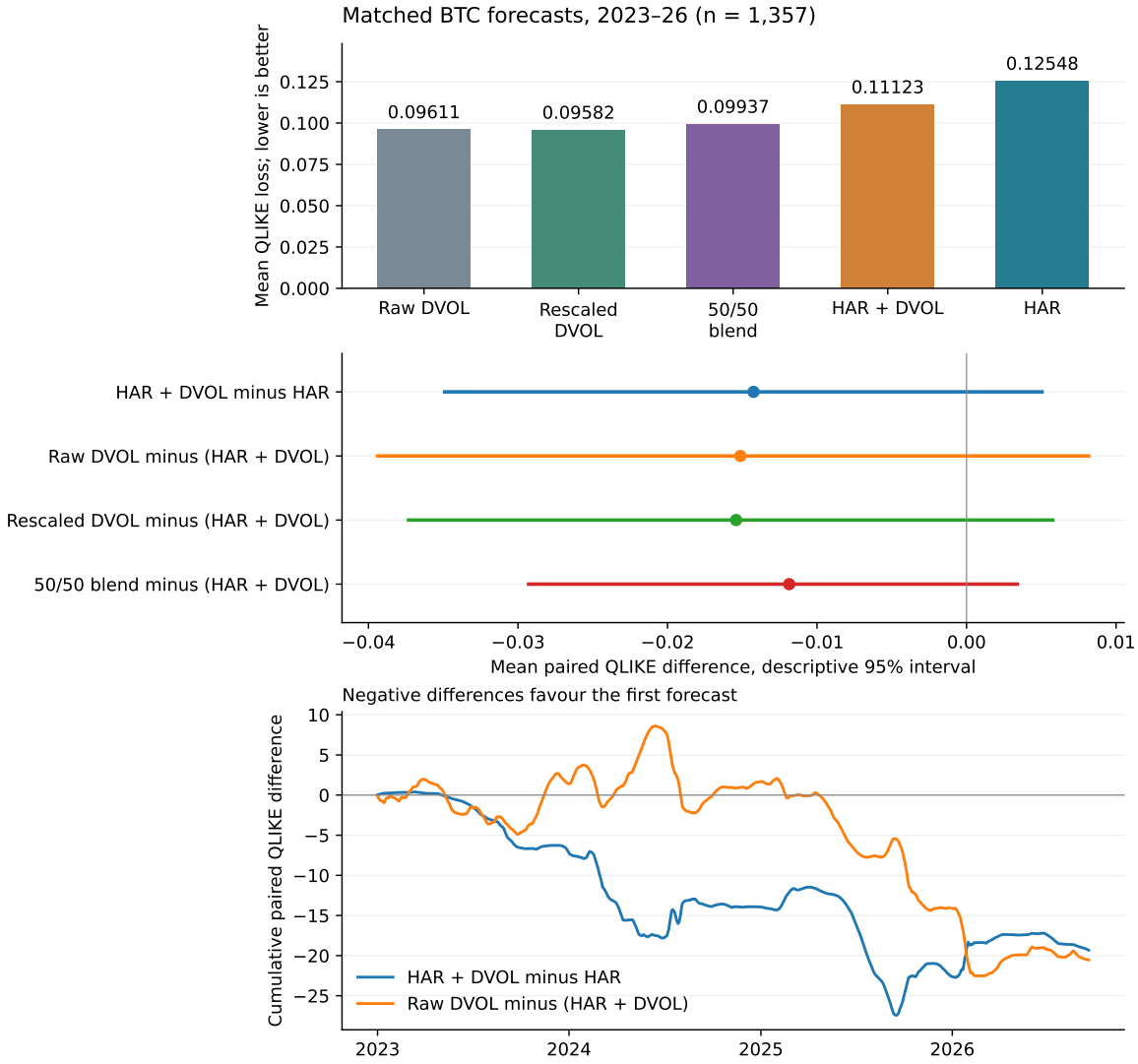


Figure 1. Mean QLIKE loss for the five matched forecasts, descriptive 95% paired intervals, and the original cumulative paired differences over 1,357 daily origins. Negative differences favour the first forecast named. Raw-DVOL, rescaling and fixed-blend comparisons are exploratory additions made after inspection.

1 Introduction

Does option-implied variance add forecasting information beyond daily, weekly and monthly realised variance? I compare a direct 30-day HAR forecast with HAR + DVOL, which adds the previous completed daily DVOL variance. Raw DVOL provides a reference forecast.

Raw DVOL's original paired interval against HAR excludes zero, but its effective sample size (ESS) is below the reporting rule and I highlighted it after inspection. Rolling rescaling and the equal blend leave the ranking uncertain.

Prior work asks whether option-implied volatility adds information beyond historical volatility. Christensen and Prabhala (1998) find incremental information in implied volatility for the S&P 100 index using non-overlapping observations; their setting differs from the overlapping BTC targets here. Corsi (2009) combines realised-volatility components across time horizons in the HAR model, which I adapt to a direct variance target. Patton (2011) studies how imperfect volatility proxies affect forecast rankings and motivates care in choosing the loss function.

Using Deribit option trades, Hoang and Baur (2020) compare implied volatility with time-series forecasts, including HAR. They find horizon-dependent accuracy and gains from combining implied volatility with model forecasts. Their option-derived measure and shorter forecast horizons differ from the direct 30-day DVOL comparison here.

2 Data and method

The retained inputs contain 48,626 hourly BTC-PERPETUAL candles and 1,996 daily DVOL candles, downloaded on 19 October 2026. Evaluation uses daily midnight forecast origins from 1 January 2023 through 18 September 2026. Each target is the next 720 hourly log-return squares, annualised by 8,760; that window requires 721 completed closes.

Both regressions use exactly the latest 365 labels whose full 30-day targets have ended. Prior 1-, 7- and 30-day annualised variances form the HAR features. HAR + DVOL adds the previous completed daily DVOL close divided by 100 and then squared. Level ordinary least squares includes an intercept, with features centred and scaled using training rows only. An annual-variance floor of 1e-6 applies to forecasts; raw predictions and every floor hit are retained. Neither regression reaches that floor in this sample. Level OLS minimises squared variance errors, while evaluation uses QLIKE. These are different objectives within the restricted HAR specifications, although both losses target conditional mean variance when forecasts are unrestricted (Patton, 2011).

With r_t denoting an hourly log return and t the forecast origin, the future target and completed-return features are

$$V_t^{(30\mathrm{d})} = \frac{8760}{720}\sum_{h=1}^{720} r_{t+h}^2,$$

$$RV_t^{(k)} = \frac{8760}{24\,k}\sum_{h=1}^{24\,k} r_{t-h+1}^2, \quad k \in \{1, 7, 30\}.$$

The two direct forecasts are

$$\widehat{V}_t^{(30\mathrm{d}),\mathrm{HAR}} = \widehat{\beta}_{0,t} + \widehat{\beta}_{d,t} RV_t^{(1)} + \widehat{\beta}_{w,t} RV_t^{(7)} + \widehat{\beta}_{m,t} RV_t^{(30)},$$

$$\widehat{V}_t^{(30\mathrm{d}),\mathrm{HAR+DVOL}} = \widehat{\gamma}_{0,t} + \widehat{\gamma}_{d,t} RV_t^{(1)} + \widehat{\gamma}_{w,t} RV_t^{(7)} + \widehat{\gamma}_{m,t} RV_t^{(30)} + \widehat{\gamma}_{D,t}\left(\frac{\mathrm{DVOL}_t^{\mathrm{prev.close}}}{100}\right)^2.$$

All comparisons use identical origins. Smaller values of the QLIKE loss indicate more accurate variance forecasts:

$$\mathrm{QLIKE} = \frac{\text{realised}}{\text{forecast}} - \log\left(\frac{\text{realised}}{\text{forecast}}\right) - 1.$$

3 Testing scale and fixed-weight alternatives

The original HAR + DVOL minus HAR mean QLIKE difference is -0.01424, with descriptive interval [-0.03490, +0.00505] and ESS 51.71. Raw DVOL minus (HAR + DVOL) has mean -0.01513, interval [-0.03939, +0.00817] and ESS 53.44.

I declared the added specifications in public commit `6a6b59b` before evaluating these additions, after inspecting the original ranking. Rescaled DVOL uses exactly the same latest 365 fully completed labels as each HAR fit. With D_t denoting raw DVOL variance, V_s the realised target and $\mathcal{T}_t$ that training set, the added forecasts are

$$c_t = \frac{1}{365}\sum_{s\in\mathcal{T}_t}\frac{V_s}{D_s}, \qquad \widehat{V}_t^{\mathrm{rescaled}} = c_t D_t, \qquad \widehat{V}_t^{\mathrm{equal}} = 0.5\widehat{V}_t^{\mathrm{HAR}} + 0.5 D_t.$$

The multiplier is positive and unclipped. The blend averages variances and has no fitted combination weight; HAR's coefficients remain estimated. All rows use the same origins, bootstrap draws and ESS machinery, while the locked prospective primary comparison stays unchanged.

Table 1. Matched exploratory comparisons against HAR + DVOL. Lower mean QLIKE is better. Negative paired differences favour the row forecast. All intervals are descriptive; the reference row has no paired interval or ESS.

Forecast	Mean QLIKE	Paired mean difference	95% interval	ESS
Raw DVOL	0.09611	-0.01513	[-0.03939, +0.00817]	53.44
Rescaled DVOL	0.09582	-0.01541	[-0.03732, +0.00577]	70.63
50/50 HAR–DVOL	0.09937	-0.01186	[-0.02929, +0.00339]	58.57
HAR + DVOL	0.11123	—	—	—
HAR	0.12548	+0.01424	[-0.00505, +0.03490]	51.71

Against HAR + DVOL, rescaled DVOL's ESS is 70.63, above the 60 reporting rule, but its interval includes zero; the blend's 58.57 is just below. Rescaled DVOL minus raw DVOL has mean -0.00028 and interval [-0.01600, +0.01550]. The equal blend minus raw DVOL has mean +0.00327 and interval [-0.01125, +0.01952]. Rescaling barely changes the mean loss; the blend has lower mean loss than HAR + DVOL but higher mean loss than raw DVOL. All four rescaling and blend intervals include zero. The simpler rules therefore leave the cause of the original ranking unresolved.

Raw DVOL minus HAR has mean -0.02937, descriptive 95% interval [-0.05831, -0.00465] and ESS 47.79. The original pipeline already computed this contrast, but I chose to highlight it after seeing the ranking, so this is a post-inspection highlight rather than a pre-declared test. Its interval excludes zero, but ESS is below the preset reporting rule of 60, and the synthetic stress tests in Section 4 show ungated null rejection of 8.5% to 45.0% at $n = 1{,}357$ against a nominal 5%. I therefore treat it as descriptive.

QLIKE penalises a forecast too low by a factor $k > 1$ more than one too high by the same factor. I therefore check scale. Mean realised-to-forecast variance ratios are 1.1689 for trailing variance over the previous 30 days, 0.8920 for raw DVOL, 0.9499 for HAR and 1.0072 for HAR + DVOL. Each averages individual $V_t/\widehat{V}_t$ ratios, rather than dividing sample means. A single scalar equal to that mean minimises in-sample QLIKE. It lowers raw DVOL's in-sample mean from 0.09611 to 0.08981 and HAR + DVOL's to 0.11121.

The completed-label rolling rule reaches 0.09582, so most of the full-sample scaling gain is absent from the rolling forecast. This gap is consistent with changes in optimal scale or rolling estimation noise; it does not identify scale drift. Even after full-sample rescaling, raw DVOL's mean loss of 0.08981 remains below unscaled HAR + DVOL's 0.11123. These in-sample diagnostics lean against helpful constant upward scaling as the explanation for raw DVOL's advantage; they establish neither conditional calibration nor a causal explanation.

3.1 Extreme differences pull against the mean

The largest 1% and 5% of absolute HAR + DVOL minus HAR QLIKE differences account for 8.6% and 27.8% of total absolute differences. Their aggregate signed contributions are positive, so these groups hurt HAR + DVOL relative to HAR even though the full-sample mean is negative. This pattern runs against a simple explanation in which HAR + DVOL wins by avoiding the most extreme losses.

Table 2. Mean paired QLIKE differences within each year. Negative values favour the first forecast named. The final year ends on 18 September; these are descriptive episode summaries.

Year	Origins	HAR + DVOL minus HAR	Raw DVOL minus (HAR + DVOL)
2023	365	-0.01943	+0.00386
2024	366	-0.01870	+0.00072
2025	365	-0.02382	-0.04327
2026 (partial)	261	+0.01265	-0.02455

Raw DVOL's advantage over HAR + DVOL is concentrated in 2025–2026. The 2025 signed loss sum accounts for 76.93% of the full-sample net advantage; the small positive differences in 2023–2024 partly offset later gains. HAR + DVOL lowers annual mean QLIKE relative to HAR in 2023–2025 and raises it in the partial 2026 sample. Leave-one-year-out diagnostics remain in the results JSON.

The retained diagnostics cover every fixed calendar quarter, extreme-origin dates and all 30 non-overlapping starting phases. Each phase contains 45 or 46 outcomes. These displays describe concentration; their interpretation is set out in the limitations below.

3.2 Lower mean QLIKE loss brings no clear budget gain

I use a variance-budget diagnostic with an illustrative annual variance target of 0.04, equivalent to 20% annual volatility. The rule scales assumed fractional exposure; its capped weight and squared variance-budget error are

$$w = \min\left(1, \frac{0.20}{\sqrt{\text{forecast variance}}}\right), \qquad \left(w^2 \times \text{realised variance} - 0.04\right)^2.$$

Table 3. The variance-budget diagnostic for the original forecasts. Budget exceeded is the share of origins above the target variance budget. Trailing variance, a naive previous-30-day realised-variance forecast with mean QLIKE 0.16670, serves as a reference in this diagnostic. The rescaled and blended forecasts were not evaluated under this diagnostic.

Forecast	Mean weight	Mean squared variance-budget error	Budget exceeded
trailing variance	0.469	0.0008186	46.7%
Raw DVOL	0.418	0.0002872	29.6%
HAR	0.420	0.0004014	36.3%
HAR + DVOL	0.437	0.0004196	41.7%

HAR + DVOL has a higher mean squared variance-budget error than HAR: the paired difference is +0.0000182, with interval [-0.0000818, +0.0001221]. The interval includes zero, so this diagnostic leaves the benefit uncertain.

3.3 The ranking depends on the specification

After inspecting v1 and the review, I publicly declared v2 in commit `0615ec0` and amended its focal contrast in `ad08a14` before evaluation. A strict window check uses 1,208 common origins; the broader 17-model comparison uses a later 1,176-origin cohort so every model has sufficient completed training labels. Its means are not directly comparable with Table 1.

The additions include log-HAR, positive log-link QLIKE fits with and without ridge shrinkage, longer and expanding windows, EWMA, GARCH, EGARCH, a constrained Bates–Granger-style combination, and Mincer–Zarnowitz-corrected DVOL. Log and QLIKE fits use 730 completed labels. Training-only residual smearing back-transforms log fits (Duan, 1983); it does not establish conditional unbiasedness. Combination weights use completed reconstructed past-only HAR forecasts, not fitted values or original unrevised vintages. The declaration fixes all settings and failure rules.

Log-HAR + DVOL has mean QLIKE 0.09278, versus 0.10888 for log-HAR and 0.09425 for raw DVOL. QLIKE-fitted HAR + DVOL scores 0.09485 and QLIKE-fitted HAR scores 0.11046: their focal difference is -0.01561, bootstrap interval [-0.03517, +0.00359], ESS 56.85. Longer-window results and all model rows appear in the complete results. These exploratory changes show sensitivity to specification; they do not identify estimation noise as the cause of the original ranking.

4 Inference limits

Paired stationary-bootstrap inference uses 5,000 draws, a fixed seed and expected blocks of 60 days, with a 59-lag heteroskedasticity- and autocorrelation-consistent (HAC) cross-check. Block sensitivities use 30, 90, 180 and 365 days. Overlapping targets and rolling fits can create dependence beyond the 30-day horizon. For each paired QLIKE difference series, ESS is the minimum of the observation count, an estimate from positive autocorrelation pairs constrained to decrease, and a variance-ratio estimate using the 59-lag Newey–West long-run variance; the pair sequence stops at its first nonpositive pair. The existing plan fixes 60 as a minimum reporting rule to withhold confirmatory results when estimated information is limited; I retain it without tuning it to these results, but the study provides no size or power derivation that makes this cutoff sufficient for valid inference.

Each declared synthetic scenario has 200 independent replications at sample sizes 365 and 1,357, with 2,000 bootstrap draws per replication. The population null has exactly zero mean, without centring each realised sample. The scenarios use a 30-day moving-average loss component alone, add a stationary slow component of persistence $\exp(-1/365)$, or add symmetric rare shocks spread across overlapping windows. The alternative shifts the mean by -0.25 marginal standard deviations, rather than -0.25 QLIKE.

Table 4. Synthetic null rejection rates before the ESS rule. Monte Carlo (MC) intervals describe simulation uncertainty; the last column is the share of replications passing the rule.

n	Synthetic loss process	Ungated rejection	95% MC interval	ESS rule passes
365	30-day overlap	19.0%	14.2% to 25.0%	0.0%
365	30-day overlap plus slow component	49.0%	42.2% to 55.9%	0.0%
365	30-day overlap plus clustered tail shocks	16.0%	11.6% to 21.7%	0.0%
1,357	30-day overlap	10.0%	6.6% to 14.9%	1.5%
1,357	30-day overlap plus slow component	45.0%	38.3% to 51.9%	0.0%
1,357	30-day overlap plus clustered tail shocks	8.5%	5.4% to 13.2%	1.5%

At $n = 365$, ungated null rejection ranges from 16% to 49%, against nominal 5%. The ESS rule passes zero of 200 replications under every one-year null and alternative. The Wilson 95% upper bound is around 1.9% for the probability of passing the rule in those scenarios. The rule therefore prevents a formal significance test in every declared one-year replication. At $n = 1{,}357$, some conditional estimates use only one to four passing replications, with wide Monte Carlo intervals.

The v2 reference calculations add the Harvey–Leybourne–Newbold (HLN) correction to the Diebold–Mariano statistic (Harvey, Leybourne and Newbold, 1997) and a declared overlapping-block subsampling candidate (Politis and Romano, 1992). On the original cohort, raw DVOL minus HAR has HLN reference p-values near the review's values, but its subsampling interval includes zero. The focal v2 subsampling interval [-0.03368, -0.00158] excludes zero while its bootstrap interval includes zero. The candidate subsampling rule fails the synthetic null-size checks, so that disagreement provides no basis for selecting it. All rejection rates, conditional block-resampled power curves and Monte Carlo intervals are retained in the results. They describe assumptions fitted to an inspected loss series, not the power of a future market sample. The exploratory model confidence set (Hansen, Lunde and Nason, 2011) retains 11 models; inclusion does not establish equal accuracy.

5 Limitations

Research choices. I added the historical comparisons, including v2, after inspecting the data. V2 was declared before its evaluation on that same inspected history. Chronological training does not remove that exposure to research choices. All historical findings and generated-forecast diagnostics are exploratory; the intervals are descriptive, and additional diagnostics are not primary discoveries. The mean QLIKE ranking establishes no formal superiority. I chose to report the raw DVOL minus HAR comparison after inspection; the raw DVOL minus (HAR + DVOL), rescaling and fixed-blend comparisons also follow inspection. A one-parameter rolling fit still has estimation risk because its overlapping labels are dependent. The added intervals do not turn these mechanism checks into confirmatory tests.

Data. The original source check compares open, high, low and close (OHLC) fields. It fails because one unused DVOL open differs. All three audited DVOL closes match, but those dates cover little of the full history. I cannot verify the original unrevised data vintages from these historical downloads. One perpetual contract supplies a proxy for spot-index returns, so the study covers one asset and venue.

Scope. I evaluate variance forecasts and a variance-budget diagnostic. The diagnostic assumes constant fractional exposure; it does not model a fixed-unit portfolio, rebalancing, funding or profit. I have not established that the rule minimises squared variance-budget error. I make no claims about net trading returns, a causal variance premium or an executable edge. The cumulative figure tracks QLIKE differences, rather than portfolio returns.

Inference. The non-overlapping phase samples can still be dependent because volatility and rolling parameter estimates persist. All starting phases reuse the same history; favourable counts are a dependent stability display, with no binomial interpretation or ex-ante regime rule. ESS estimates dependence-adjusted information rather than counting independent episodes. The synthetic stress tests approximate loss dependence, not the full rolling HAR estimation pipeline. None of the declared one-year replications passes the ESS rule, but zero observed passes does not mean the true probability is zero; the Wilson bound quantifies that uncertainty.

6 Prospective protocol

The primary prospective contrast remains HAR + DVOL minus HAR. Under the inference-validation amendment, I treat the planned year as timestamped evidence collection without a formal significance test. I fix the cohort and terminal date in advance, even if the result is inconclusive; a later inference design needs a separate prospective declaration.

The HAR cohort contains 365 scheduled origins from 21 October 2026 through 20 October 2027, each at 01:00 UTC. Features end at midnight, forecasts freeze between 00:10 and 00:50, and outcomes begin at 01:00. Training uses the correspondingly shifted historical 30-day targets completed by midnight, a different alignment from the retrospective midnight targets. Final maturity is 19 November 2027, with terminal analysis fixed at 21 November 2027, 01:00 UTC.

Each new forecast requires matching hourly and daily DVOL closes and completion of fitting and the durable write before the freeze deadline. Missed scheduled days count as missed and stay unfilled. Terminal intervals require all 365 planned pairs and source checks; the fixed analysis cutoff excludes later-arriving scores.

A Reproduction

The README lists the exact Python environment and offline commands: `python scripts/build_research.py` rebuilds v1; `python scripts/build_review.py` rebuilds v2 and checks that the v1 summary is unchanged. Numeric manuscript macros are generated from the two summary JSON files and checked in CI. The methods note holds collection details and operational limits; the claim map links evidence to code and tests.

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